

Pellevoisin: the Raspberry Pi 4 header

Pin 1 top left with the board's edge at the top: odd pins in the inner row, even pins in the outer row next to the edge. Highlighted cells are this organ's connections. Every GND pin is the Pi's ground, and nothing on the organ's side ever touches this header.

ours	function	pin	pin	function	ours
MIDI: 6N137 module IN+	3.3 V	1	2	5 V	free (Pi fed by USB-C)
free; I2C stays off	GPIO 2 / SDA	3	4	5 V	free
ON/OFF SWITCH, one side	GPIO 3 / SCL	5	6	GND	ON/OFF SWITCH, other side
	GPIO 4	7	8	GPIO 14 / TXD	MIDI: 6N137 module IN-
RESET opto cathode (PC817 B pin 2)	GND	9	10	GPIO 15 / RXD	free
RESET: 330 ohm to PC817 B anode	GPIO 17	11	12	GPIO 18	
	GPIO 27	13	14	GND	
	GPIO 22	15	16	GPIO 23	
	3.3 V	17	18	GPIO 24	PUMP: SSR terminal 3 (+)
	GPIO 10	19	20	GND	PUMP: SSR terminal 4 (-)
	GPIO 9	21	22	GPIO 25	12 V ON: 330 ohm to PC817 A anode
	GPIO 11	23	24	GPIO 8	
12 V ON opto cathode (PC817 A pin 2)	GND	25	26	GPIO 7	
never	ID_SD	27	28	ID_SC	never
	GPIO 5	29	30	GND	
	GPIO 6	31	32	GPIO 12	
	GPIO 13	33	34	GND	
	GPIO 19	35	36	GPIO 16	
	GPIO 26	37	38	GPIO 20	
	GND	39	40	GPIO 21	

Each line is paired with its nearest ground so every cable is a short twisted pair: 5 and 6 for the switch, 11 and 9 for reset, 18 and 20 for the pump, 22 and 25 for the 12 V. The MIDI pair is pins 1 and 8 and carries no ground: TX is its return. Pins 3 and 5 are the I2C bus, which stays disabled so GPIO 3 is free for the switch and wakes a halted Pi when pulled low. The DSI screen uses its own ribbon connector.

Part pinouts: PC817 pin 1 anode, 2 cathode, 3 emitter, 4 collector (dot at pin 1). SSR-xx DA: 1 and 2 load (mains), 3 positive and 4 negative input, 3 to 32 V DC. 6N137 module: IN+ and IN- marked on the terminals; VCC, OUT, GND on the other side. organ_web: --pump 24 --power 25 --power-switch 3; reset on 17.

Pellevoisin: the box, wired

Two halves in one aluminium case, and exactly four things cross between them, each by light: the MIDI opto, the reset opto, the PS_ON opto and the SSR.

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MAINS -- IEC inlet with fuse +- E ---- earth stud on the case ---- motor frame
|
+- N -----+-----+----- motor N
|
+- L -- PANEL SWITCH, 2 poles, 3 positions, centre off, motor rated
      pole A common +- BOOK -----> motor L
                      +- AUTO -> SSR 1 -- SSR 2 -----> motor L
      pole B common --- AUTO -> ATX L and Pi PSU L
      (OFF: neither pole connected; the ATX fan cools the box in Auto)
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ORGAN SIDE (the ATX's ground)	PI SIDE (the Pi PSU's ground)
ATX 12 V, several yellows -> 12 V bus bar	Pi PSU 5 V --USB-C--> Pi 4
ATX GND, several blacks --> organ GND bus bar	DSI ribbon --> touchscreen
12 V bus / GND bus -----> boards 1-4, 5 A fuses	GPIO 3 (pin 5) -- antique switch -- Pi GND on/off
last board ISP pin 2 -----> RJ12 pin 6 = 5 V bus	GPIO 14 TX (pin 8) -- 6N137 IN- MIDI
RJ12 pin 1 = /RESET bus (jumped board to board)	3.3 V (pin 1) ----- 6N137 IN+
RJ12: MIDI signal, GND, as shipped	GPIO 24 (pin 18) -- SSR 3 (+) pump
	Pi GND ----- SSR 4 (-)
6N137 module: VCC < RJ12 pin 6, GND < RJ12 GND,	GPIO 25 (pin 22) -[330 ohm]- PC817 A anode 12 V on
OUT > boards' MIDI in	Pi GND ----- PC817 A cathode
PC817 A: collector - ATX PS_ON (green), emitter - ATX GND	GPIO 17 (pin 11) -[330 ohm]- PC817 B anode reset
PC817 B: collector - RJ12 pin 1, emitter - RJ12 GND	Pi GND ----- PC817 B cathode
SSR 1 / 2: on the mains side, above; MOV across them	

Rules

- No wire, bus bar or chassis lug between the organ's GND bus and the Pi's ground. The earth stud carries earth only; the two supplies earth themselves through their own mains leads, and neither DC negative goes to the stud.
- Mains bus bars shrouded, or covered DIN terminal blocks instead. Low-voltage bars may be bare.
- Mains and low voltage on opposite sides of a partition: inlet, switch, supplies' inputs, SSR 1 and 2, motor lead on one side; Pi, RJ12 bus, optos, SSR 3 and 4 on the other. A wire that must cross does so once, at right angles.
- The SSR's heat sink in the ATX fan's stream, thermal paste under the module. A varistor across SSR 1 and 2.
- Only the live is switched, ever. Neutral and earth run straight through. The panel switch is motor rated: 1 HP at 125 V, or 20 A. The antique on/off switch carries microamps and may be anything.
- Book: the motor runs, nothing else has power. Auto: the box comes alive; the ATX waits on standby until the Pi pulls PS_ON; the motor runs only when the SSR says so. Off: everything isolated (the SSR alone leaks a milliamp).
- Before the panel goes from Auto to Off: the antique switch off, or Shut down on the Service tab, and wait for a dark screen.